

Harbor Goods: migration to AWS

Fictional sample. Harbor Goods and all data here are fictional. Each repository in this portfolio is a separate engagement with Harbor Goods, a fictional mid-size retailer. Account IDs are AWS documentation examples. The repository files and the evidence regenerated by `make evidence` provide every number in this document.

Item	Value
Client	Harbor Goods, a fictional mid-size retailer
Scope	Warehouse and inventory application: 12 virtual machines in a colocation data center; wave 1 in full, waves 2 and 3 at plan level
Target	AWS, <code>us-east-1</code> , one production account (111122223333)
Deliverables	Inventory, 7R classification, wave plan, Terraform, DMS tasks, cutover runbook, acceptance criteria
Revision	1.0

Executive summary

Harbor Goods’ storefront already runs on AWS; its warehouse and inventory application still runs in a colocation data center. The plan moves that application to AWS in a four-hour Sunday window. Its acceptance targets are a write pause of no more than 30 minutes for warehouse staff and for the storefront’s inventory calls, and, during the seven days that follow, a rollback to the data center at any time with zero lost fulfillment orders. These are targets for the rehearsal and the cutover to meet, not measured results.

The nine risks identified during discovery divide equally into high, medium and low severity; the complete register appears under “Risks.” Each of the three high-severity issues would block or break cutover if missed.

Severity	Count	Examples
High	3	BI reads the replica that wave 1 retires; a table without a primary key; the parcel carrier allowlists the source IP
Medium	3	Nightly jobs inside the window; one-hour DNS TTL; PostgreSQL 14 to 16 during the move
Low	3	No WAF yet; application secret without rotation; DMS TLS without certificate checks

The three priority recommendations are:

1. Request the parcel carrier’s allowlisting of the new egress addresses now (prerequisite P-04). This prerequisite takes the longest to arrange, and go-live cannot proceed without it.
2. Approve the proposed first wave, moving the warehouse web app and inventory API to ECS Fargate and replatforming the database onto RDS for PostgreSQL 16 Multi-AZ. Use AWS DMS for full load followed by change data capture, with a reverse task supporting rollback.

3. Complete the full rehearsal (P-09); keep the scheduled date only if the freeze takes no more than 27 minutes (G-04).

Scope and method

`docs/methodology.md` defines the sequence: discovery, 7 Rs classification, dependency-based waves, sizing from measurements, data migration planning, a runbook expressed as data and a target built as code. Automated checks cover every step. E-06 records the plan checks, E-07 records the runbook checks, and E-08 contains the Terraform test results.

Application code changes and a cost estimate are excluded; a real engagement would include an AWS Pricing Calculator export. Detailed runbooks for waves 2 and 3 are also outside this scope and will be written after wave 1 exits.

Current state

The inventory (E-01) lists eight applications across twelve virtual machines. An HAProxy pair fronts two warehouse web app servers and two inventory API servers. Warehouse staff and handheld scanners use the web app; the storefront on AWS calls the inventory API for stock levels and sends it fulfillment orders. PostgreSQL 14 runs on a primary and streaming replica, holding 182 GB across 55 tables in five schemas. The remaining servers host nightly batch jobs, a self-hosted BI tool, supplier SFTP and a rarely used intranet wiki. Both the web app and the API tolerate a 30-minute write freeze and require recovery within 30 minutes (RTO), with zero data loss (RPO 0).

7R classification

Strategy	Servers	Why
Replatform	SRV-01 to SRV-07	Stateless web and API tiers run as containers unchanged; a managed load balancer and database remove hand patching
Retire	SRV-08, SRV-12	Multi-AZ replaces the replica; the wiki has no edits in the discovery sample
Rehost	SRV-09	Batch jobs depend on local paths; containerizing them needs code changes
Repurchase	SRV-10	14 dashboards move to a managed BI service
Retain	SRV-11	Supplier contracts allowlist the SFTP address

For each server, `plan/classification.yaml` and E-02 record the selected target, the reason for choosing it and the alternative rejected. Refactoring is not assigned to any server because no business driver justifies that rework as part of the migration (ADR 0001).

Wave plan

Wave	Servers	Exit criterion
1 Warehouse core	SRV-01 to SRV-08	All acceptance criteria met after seven days of hypercare
2 Batch and intranet	SRV-09, SRV-12	Three nightly runs match the on-premises row counts

Wave	Servers	Exit criterion
3 Reporting	SRV-10	Finance signs off the rebuilt dashboards after one month-end close
Retained	SRV-11	Revisit when supplier contracts renew

Between waves, three connections use the Site-to-Site VPN (E-03). After wave 1, batch and BI connect to RDS; after wave 2, batch connects to the retained SFTP server. Database access through Terraform is limited to the batch and BI hosts, using the /32 addresses specified in the plan.

Target architecture

Harbor Goods wave 1: target architecture and cutover path

Route 53 weighted records move users from the data center to ECS and RDS; AWS DMS keeps both databases in step (fictional)

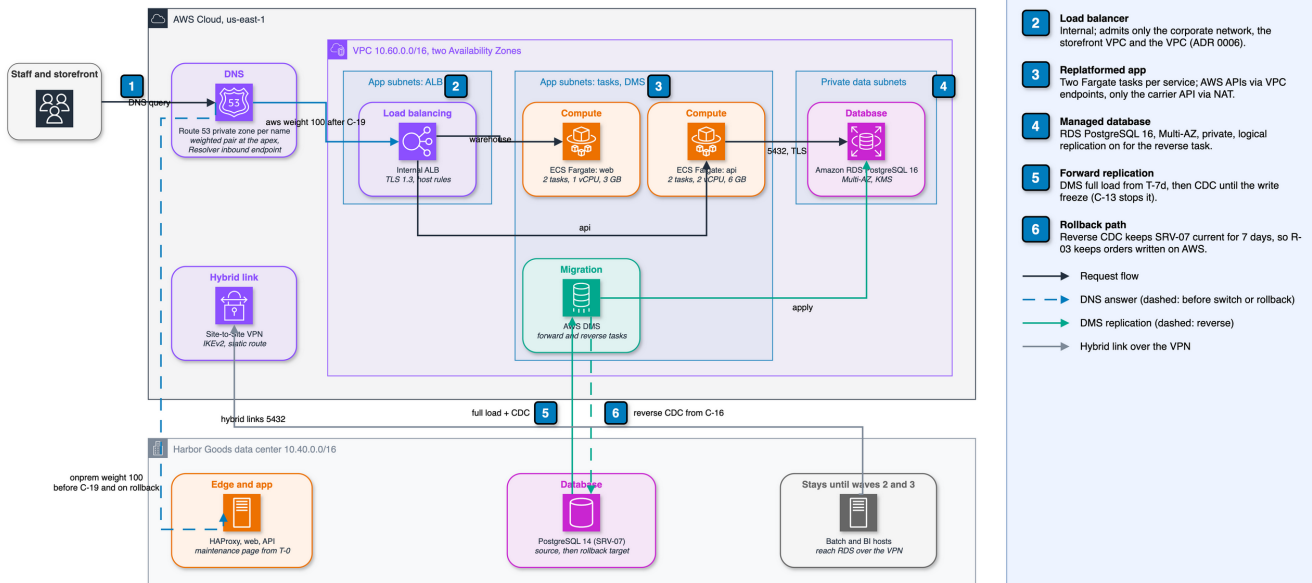


Figure 1: Wave 1 target architecture and cutover path

The application is internal, and the target keeps it that way (ADR 0006). Warehouse staff arrive from the corporate network over the Site-to-Site VPN and the storefront from its own VPC over a peering connection. Both names resolve from Route 53 weighted records, each name in its own private hosted zone so the rest of **example.com** still resolves publicly; the corporate DNS servers forward them to a Resolver inbound endpoint, and the public zone keeps only the ACM validation records. Using a TLS 1.3 policy, an internal Application Load Balancer in the private application subnets directs each host name to its ECS Fargate service. Its security group admits only the corporate network, the storefront VPC and the VPC itself, and its access logs go to the central log archive bucket, because ALB access logging supports only SSE-S3. The tasks reach AWS APIs through VPC endpoints and send nothing to the internet except calls to the parcel carrier's published addresses, which leave through the NAT gateways the carrier allowlists. The RDS for PostgreSQL 16 database uses Multi-AZ and a customer managed KMS key for encryption; its data subnets have no internet route. From the application subnets, AWS DMS connects to the data center through the VPN, to RDS within the VPC and to Secrets Manager through its endpoint.

Discovery p95 measurements determine sizing, with 25 percent headroom (E-04). Each inventory API task has 2 vCPU and 6 GB, and each web app task has 1 vCPU and 3 GB. The database uses a **db.r7g.2xlarge** instance and 400 GB of gp3 storage.

Data migration

At T-7d, the forward task begins a full load of all five schemas, then streams changes up to the write freeze (E-05). DMS cannot keep `audit.request_log` synchronized because the table lacks a primary key and its retention job deletes rows. A `pg_dump` export transfers its history at T-24h; `\copy` transfers the remaining rows during the freeze. Manual steps M-01 to M-05 handle sequences, roles and grants, which DMS does not copy. When cutover occurs, a reverse task begins streaming changes from RDS to the data center, allowing rollback without data loss after go-live (ADR 0002).

Cutover and rollback

The runbook defines 13 prerequisites and 23 timed steps, with 8 go/no-go criteria distributed across three checkpoints. It also includes 7 validation queries, 7 rollback triggers and 3 rollback procedures:

Phase	Window	Rollback	Time to roll back
Preparation	T-72h to T-0	R-01: abort, nothing to undo	15 min
Write freeze	T-0 to T+27m	R-02: restart the data center application; RDS is reloaded before the next attempt	15 min
Hypercare	T+27m to T+7d	R-03: maintenance page on AWS, drain the reverse task, flip DNS back	25 min

For both DNS names, the switch sets Route 53 weights to 0/100 from 100/0 in the private hosted zones. Because the database permits only one writer, the Terraform module rejects intermediate weights (ADR 0003). Three days before the switch, the TTL of the public records is reduced from 3600 seconds to 60 seconds; two days before, the corporate DNS servers start forwarding both names to the private zones and the public records are deleted. Fixture-based execution on PostgreSQL 14 and 16 verified the validation queries (`make sql-check`).

Acceptance criteria

Acceptance of wave 1 requires seven days of hypercare and satisfaction of all ten conditions in `runbooks/acceptance-criteria.m`. These include a 99.9 percent pass rate for synthetic checks, inventory API p95 no higher than 350 ms, an actual write freeze of no more than 30 minutes, zero lost fulfillment orders and completion of a point-in-time restore within 60 minutes.

Risks

ID	Severity	Risk	Evidence	Treatment
RISK-01	High	BI dashboards read the replica that wave 1 retires	E-03	Repoint BI to RDS over the VPN in wave 1 (C-20)
RISK-02	High	<code>audit.request_log</code> has no primary key and a delete job	E-05	Exclude from DMS; export the history and copy the tail in the freeze (M-05)
RISK-03	High	The parcel carrier allowlists the data center egress address	E-01	Register the NAT gateway addresses before cutover (P-04)

ID	Severity	Risk	Evidence	Treatment
RISK-04	Medium	JOB-01 and JOB-03 run inside the window against the database	E-07	Disable at T-80m (C-06), final dump at C-07, re-enable batch after the switch
RISK-05	Medium	One-hour TTL on the public names	E-07	Lower to 60 s at T-72h (C-01); delete them once corporate DNS forwards to the private zone (C-02)
RISK-06	Medium	PostgreSQL 14 to 16 during the move	E-04	Full rehearsal on 16 (P-09); smoke tests before the switch
RISK-07	Low	No AWS WAF in front of the internal load balancer (none on premises either)	E-08	Add managed rules in count mode during hypercare, then block
RISK-08	Low	The application database secret has no rotation	E-08	Enable rotation once the API reloads credentials, after hypercare
RISK-09	Low	DMS endpoints use TLS without certificate verification	E-08	Import the RDS and data center CAs and switch to verify-full

Limits

- All material is fictional and synthetic, so none of the findings apply to an actual system.
- Offline testing establishes consistency among the plan, runbook and Terraform, and confirms Terraform's intended properties. It does not establish that AWS accepts every setting. The manually run `make test-live` checks the network and database slice in a sandbox account.
- Planning for waves 2 and 3 covers dependencies; after wave 1 exits, their runbooks will be developed using the same format.

Evidence index

ID	File	Produced by
E-01	Inventory summary	<code>scripts/build_evidence.py</code>
E-02	Classification and waves	<code>scripts/build_evidence.py</code>
E-03	Dependencies by wave	<code>scripts/build_evidence.py</code>
E-04	Target sizing	<code>scripts/build_evidence.py</code>
E-05	DMS coverage	<code>scripts/build_evidence.py</code>
E-06	Plan checks	<code>scripts/check_plan.py</code> rules
E-07	Runbook checks	<code>scripts/check_runbook.py</code> rules
E-08	Terraform tests	<code>terraform test</code> with mock providers

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